

Software Engineering

Software Quality

Who is responsible for Software Quality?

Who is responsible for Software Quality?

Testers / Quality (Assurance) Managers

Who is responsible for Software Quality?

~~Testers / Quality Assurance Managers~~

Everyone.

[“Whole Team Quality” approach](#) by Janet Gregory and Lisa Crispin

Software Engineering: Basic Terminology

Roles

Project Manager / Product Owner / Product Manager

(UI/UX) Designer

Developer / Software Engineer

Tester / Quality (Assurance) Manager / Quality Engineer

Operations Admin / Infrastructure Engineer / Site Reliability Engineer / ~~The DevOps guy~~

User

What is Software Quality?

What is Software Quality?

- Functional Suitability
- Performance Efficiency
- Compatibility
- Interaction Capability (aka Usability)
- Reliability
- Security
- Maintainability
- Flexibility
- Safety

[Source: ISO 25010](#)

What is Software Quality?

Functional Suitability

This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions. This characteristic is composed of the following sub-characteristics:

- **Functional completeness** - Degree to which the set of functions covers all the specified tasks and intended users' objectives.
- **Functional correctness** - Degree to which a product or system provides accurate results when used by intended users.
- **Functional appropriateness** - Degree to which the functions facilitate the accomplishment of specified tasks and objectives.

What is Software Quality?

Performance Efficiency

This characteristic represents the degree to which a product performs its functions within specified time and throughput parameters and is efficient in the use of resources (such as CPU, memory, storage, network devices, energy, materials...) under specified conditions. This characteristic is composed of the following sub-characteristics:

- **Time behaviour** - Degree to which the response time and throughput rates of a product or system, when performing its functions, meet requirements.
- **Resource utilization** - Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.
- **Capacity** - Degree to which the maximum limits of a product or system parameter meet requirements.

What is Software Quality?

Compatibility

Degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions while sharing the same common environment and resources. This characteristic is composed of the following sub-characteristics:

- **Co-existence** - Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.
- **Interoperability** - Degree to which a system, product or component can exchange information with other products and mutually use the information that has been exchanged.

What is Software Quality?

Interaction Capability (aka Usability)

Degree to which a product or system can be interacted with by specified users to exchange information via the user interface to complete specific tasks in a variety of contexts of use. This characteristic is composed of the following sub-characteristics:

- **Appropriateness recognizability** - Degree to which users can recognize whether a product or system is appropriate for their needs.
- **Learnability** - Degree to which the functions of a product or system can be learnt to be used by specified users within a specified amount of time.
- **Operability** - Degree to which a product or system has attributes that make it easy to operate and control.
- **User error protection** - Degree to which a system prevents users against operation errors.
- **User engagement** - Degree to which a user interface presents functions and information in an inviting and motivating manner encouraging continued interaction.
- **Inclusivity** - Degree to which a product or system can be used by people of various backgrounds (such as people of various ages, abilities, cultures, ethnicities, languages, genders, economic situations, etc.).
- **User assistance** - Degree to which a product can be used by people with the widest range of characteristics and capabilities to achieve specified goals in a specified context of use.
- **Self-descriptiveness** - Degree to which a product presents appropriate information, where needed by the user, to make its capabilities and use immediately obvious to the user without excessive interactions with a product or other resources (such as user documentation, help desks or other users).

What is Software Quality?

Reliability

Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time. This characteristic is composed of the following sub-characteristics:

- **Faultlessness** - Degree to which a system, product or component performs specified functions without fault under normal operation.
- **Availability** - Degree to which a system, product or component is operational and accessible when required for use.
- **Fault tolerance** - Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.
- **Recoverability** - Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.

What is Software Quality?

Security

Degree to which a product or system defends against attack patterns by malicious actors and protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization. This characteristic is composed of the following sub-characteristics:

- **Confidentiality** - Degree to which a product or system ensures that data are accessible only to those authorized to have access.
- **Integrity** - Degree to which a system, product or component ensures that the state of its system and data are protected from unauthorized modification or deletion either by malicious action or computer error.
- **Non-repudiation** - Degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.
- **Accountability** - Degree to which the actions of an entity can be traced uniquely to the entity.
- **Authenticity** - Degree to which the identity of a subject or resource can be proved to be the one claimed.
- **Resistance** - Degree to which the product or system sustains operations while under attack from a malicious actor.

What is Software Quality?

Maintainability

This characteristic represents the degree of effectiveness and efficiency with which a product or system can be modified to improve it, correct it or adapt it to changes in environment, and in requirements. This characteristic is composed of the following sub-characteristics:

- **Modularity** - Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.
- **Reusability** - Degree to which a product can be used as an asset in more than one system, or in building other assets.
- **Analysability** - Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.
- **Modifiability** - Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.
- **Testability** - Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.

What is Software Quality?

Flexibility

Degree to which a product can be adapted to changes in its requirements, contexts of use or system environment. This characteristic is composed of the following sub-characteristics:

- **Adaptability** - Degree to which a product or system can effectively and efficiently be adapted for or transferred to different hardware, software or other operational or usage environments.
- **Scalability** - Degree to which a product can handle growing or shrinking workloads or to adapt its capacity to handle variability.
- **Installability** - Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.
- **Replaceability** - Degree to which a product can replace another specified software product for the same purpose in the same environment.

What is Software Quality?

Safety

This characteristic represents the degree to which a product under defined conditions to avoid a state in which human life, health, property, or the environment is endangered. This characteristic is composed of the following sub-characteristics:

- **Operational constraint** - Degree to which a product or system constrains its operation to within safe parameters or states when encountering operational hazard.
- **Risk identification** - Degree to which a product can identify a course of events or operations that can expose life, property or environment to unacceptable risk.
- **Fail safe** - Degree to which a product can automatically place itself in a safe operating mode, or to revert to a safe condition in the event of a failure.
- **Hazard warning** - Degree to which a product or system provides warnings of unacceptable risks to operations or internal controls so that they can react in sufficient time to sustain safe operations.
- **Safe integration** - Degree to which a product can maintain safety during and after integration with one or more components.

**How can you agree on the level of
Software Quality needed?**

How can you agree on the level of Software Quality needed?

Problems when over engineering:

- More complexity than needed slows the development of new features down
- More resources than needed increase the cost for infrastructure

Problems when neglecting quality:

- Unsuitable system architecture decisions are hard to change
- Allocation of user facing errors damage the reputation
- Some unmanaged risks may only occur once but the potential damage is so severe that company will not recover from it
- Slowing down the development of new features
 - The later an error is detected in the process of Software Engineering, the more time it takes to fix it
 - The worse the quality of a software product, the more difficult it gets to add new functionality
- Further degradation of quality as a result of being used to bad quality ("Broken Window Theory")

Exercise - 15 minutes

Quality Criteria

- Look up ISO 25010 (QR Code)
- **On your own:** What are the top 3 quality criteria that the company you work(ed) for is optimising for?
- **Discuss in groups of 3-4:** Is there one quality criterion that (most of) you have in common?
- **One of you shares with the class:** which one is it and why?



Quality Criteria ISO 25010

Functional Suitability

- Functional completeness
- Functional correctness
- Functional appropriateness

Performance Efficiency

- Time behaviour
- Resource utilization
- Capacity

Compatibility

- Co-existence
- Interoperability

Interaction Capability (aka Usability)

- Appropriateness recognizability
- Learnability
- Operability
- User error protection
- User engagement
- Inclusivity
- User assistance
- Self-descriptiveness

Reliability

- Faultlessness
- Availability
- Fault tolerance
- Recoverability

Security

- Confidentiality
- Integrity
- Non-repudiation
- Accountability
- Authenticity
- Resistance

Maintainability

- Modularity
- Reusability
- Analysability
- Modifiability
- Testability

Flexibility

- Adaptability
- Scalability
- Installability
- Replaceability

Safety

- Operational constraint
- Risk identification
- Fail safe
- Hazard warning
- Safe integration

Quality Tradeoffs

“We trade Interaction Capability (aka Usability) for Security”

Not giving the user the information whether it's
their username or their password that is not correct
means ***less user assistance*** at login
and we deliberately want this as it also means
less information for a potential attacker
and therefore ***increases confidentiality***

Quality Scenarios

A remote user sends a request for a report over the internet during high load and receives it within 5 seconds

The data on the db server become unusable due to an unforeseen event and can be restored in 15 minutes

A newly joined team member receives the source code of the system and is able to understand it and contribute a new feature within a week

Quality Scenarios

A remote user sends a request for a report over the internet during high load and receives it within 5 seconds

- Performance Efficiency -> Time Behaviour

The data on the db server become unusable due to an unforeseen event and can be restored in 15 minutes

- Reliability -> Recoverability

A newly joined team member receives the source code of the system and is able to understand it and contribute a new feature within a week

- Maintainability -> Analysability & Modifiability

Quality Scenarios

A remote user sends a request for a report over the internet during high load and receives it within 5 seconds

- Performance Efficiency -> Time Behaviour

The data on the db server become unusable due to an unforeseen event and can be restored in 15 minutes

- Reliability -> Recoverability

A newly joined team member receives the source code of the system and is able to understand it and contribute a new feature within a week

- Maintainability -> Analysability & Modifiability

Exercise - 15 minutes

In pairs: come up with 2 quality scenarios inspired by your work experience

Exercise: Quality Scenarios and Criteria

Marketing created this pitch for the software you should create with your team:

MentorHive — Unlock Expertise Anytime, Anywhere

Finding the right mentor shouldn't be complicated. With MentorHive, you can instantly connect with professionals across every industry — from tech and design to finance and marketing — all ready to guide you with real-world experience.

MentorHive is the smart, streamlined platform that matches you with curated mentors based on your goals, skill level, and availability. Whether you're switching careers, leveling up your current role, or exploring a new passion, MentorHive makes expert guidance accessible in just a few clicks.

Exercise: Quality Scenarios and Criteria

What makes MentorHive different for learners?

- Pair with mentors whose background truly fits your goals.
- Video, screen-sharing, and collaborative tools built right in.
- Set milestones and track your progress as you grow.

What makes MentorHive different for mentors?

- Share your knowledge with motivated learners.
- Take calls when it fits your schedule.
- Get compensated reliably without administrative work.
- See how your mentees grow with your guidance.

Start your mentorship journey today — and turn ambition into action.

Exercise: Define two quality scenarios based on the software scenario. Reference the quality criterion according to ISO 25010 that the quality scenario refers to. You may add information that is missing.

